

BOUTHAYNA HAMMAMI

SOFTWARE ARCHITECTURE ENGINEERING STUDENT | ESPRIT

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SUMMARY

Final-year engineering student specializing in Software Architecture at ESPRIT, with solid hands-on experience in full-stack development (Angular, React, .NET, Spring Boot, FastAPI), mobile development (Flutter), AI/LLM integration and microservices architecture design.
Autonomous across the entire DevOps cycle (CI/CD, Docker, Kubernetes) and committed to software testing best practices (JUnit, pytest, Playwright, ISTQB, SonarQube).
Rigorous and curious, I am looking for a final-year internship (PFE) or a position in software architecture, full-stack development or software quality, combining strong technical design with an innovative mindset.

SKILLS

- **Architecture & Design :** Microservices architecture, layered architecture, API Gateway, Service Discovery (Eureka), Config Server, asynchronous messaging (RabbitMQ), REST API
- **Languages:** Java, Python, Dart, TypeScript, PHP, JavaScript, HTML, CSS
- **Web Frameworks:** Spring Boot, Angular, React, FastAPI, ASP.NET Core (.NET), Entity Framework Core
- **Mobile:** Flutter, Firebase Cloud Messaging, Android (Android Studio)
- **Databases:** MySQL, PostgreSQL, PLSQL, Azure SQL, MongoDB, Elasticsearch, Hive
- **DevOps & Cloud:** Docker, Kubernetes, Jenkins (CI/CD), Git / GitHub, Azure
- **Security:** Keycloak, OAuth2, JWT, OTP / email verification, secret hashing
- **Testing & Quality:** JUnit, pytest, Vitest, Playwright (E2E), Postman, SonarQube, ISTQB
- **BI, Data & AI:** Power BI, DAX, ETL, Kibana, ELK Stack, Machine Learning, pandas, LLM (Groq), prompt engineering
- **Methodologies:** Agile / Scrum

PROFESSIONAL EXPERIENCE

Terreau Project **June - August 2026**
IDA Conseil Informatique Inc.
SaaS platform connecting Tunisian talents, Quebec SMEs, project owners and investors (Angular 19, ASP.NET Core .NET 10, Azure SQL, EF Core 10).

- Developed Angular 19 interfaces (standalone components, signals) for several platform modules, following a custom design system (Angular Material + Lucide).
- Contributed to the ASP.NET Core / EF Core 10 backend (profiles, missions, application workflows) and to the multi-role architecture (guards, protected routes, authorization).
- Worked in Agile/Scrum with 2-week sprints, code reviews and regular progress meetings.

ELK Dashboard Implementation **February - May 2024**
TALYS

- Analyzed and specified requirements for a KPI visualization and filtering solution through a custom dashboard built on the ELK stack (Elasticsearch, Kibana) for multiple user profiles.
- Integrated data and automated its ingestion into Elasticsearch (performance, security, scalability); upskilled in Angular, Spring Boot and Kibana in a SCRUM environment.

ACADEMIC PROJECTS

AI Data Analyst — AI-Powered Data Analysis Assistant (FastAPI, React, LLM)

- Web app to upload files (CSV, Excel, PDF, Word, images) and query them in natural language: an LLM (Groq) generates the pandas code, results and charts.
- Secure execution of generated code (AST validation, function allow-list) and personal data masking before anything is sent to the model.
- FastAPI / PostgreSQL API (JWT, quotas, admin area), React / TypeScript frontend, tests with pytest, Vitest and Playwright.

LouageGo — Louage (Shared Taxi) Booking Mobile App (Flutter / Dart)

- Cross-platform mobile app with passenger and driver spaces: trip search on a map, seat selection, favorites, reviews and push notifications.
- Digital ticketing: QR and PDF tickets, scan and validation by the driver, seat availability check before each booking.
- Layered architecture with Riverpod, GoRouter and Hive; OTP authentication; trilingual UI (FR / EN / AR); 110+ tests.

Jungle In English — Microservices Architecture, DevOps & Machine Learning

- Modular academic and social management platform based on a microservices architecture: Spring Boot (backend), Angular (frontend), Keycloak (centralized SSO).
- CI/CD pipelines (Jenkins), Docker containerization and Kubernetes orchestration; features: learning paths, community spaces, performance dashboards.
- Integrated Machine Learning modules: recruitment/staffing optimization, student dropout risk prediction, personalized study plan generation.

Hôtel Brise Douce — Hotel Management Platform

- Hotel booking platform based on a microservices architecture.
- Centralized API Gateway, Service Discovery (Eureka) and centralized configuration (Config Server); asynchronous communication via RabbitMQ, MySQL persistence, Docker containerization.
- Secure authentication with Keycloak (OAuth2/JWT); Angular 18 frontend with Server-Side Rendering (Angular Universal).

Hospital Dashboard — Power BI, ETL & Machine Learning

- Designed an ETL pipeline and modeled a multi-table data warehouse (patients, visits, doctors, diagnoses, insurance, cities) in Power BI.
- Built a 6-page interactive dashboard (Overview, Financial, Patients, Providers, Geographic, Predictions) with dynamic navigation, dynamic filters (date, region) and 25+ DAX measures (revenue, emergency rate, satisfaction, unpaid rate, etc.).
- Integrated a Machine Learning revenue forecasting model (Predicted vs Actual) visualized directly in the report to support decision-making.

FORMATION

Engineering Degree — Software Architecture **2025 - Présent**
Private Higher School of Engineering and Technology (ESPRIT), Ariana

Bachelor's Degree in Computer Science and Multimedia **2021 - 2024**
Higher Institute of Multimedia Arts (ISAMM), Manouba

Technical Baccalaureate — Honors (Mention Bien) **2020 - 2021**
Route de Chouigui High School, Tebourba

LANGUES

- **Arabe:** Native
- **Français:** Fluent
- **Anglais:** Fluent
- **Allemand:** Basic